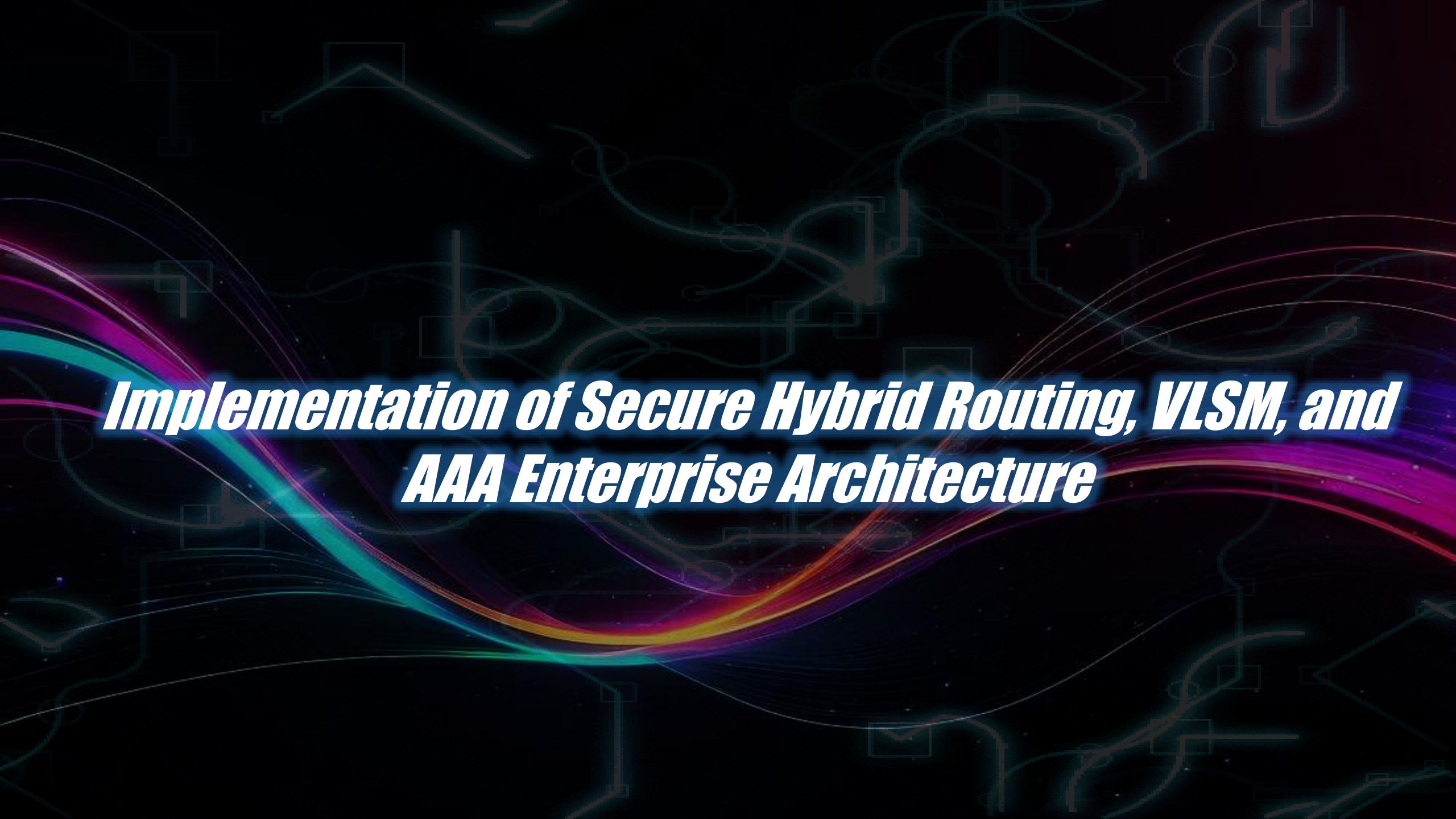


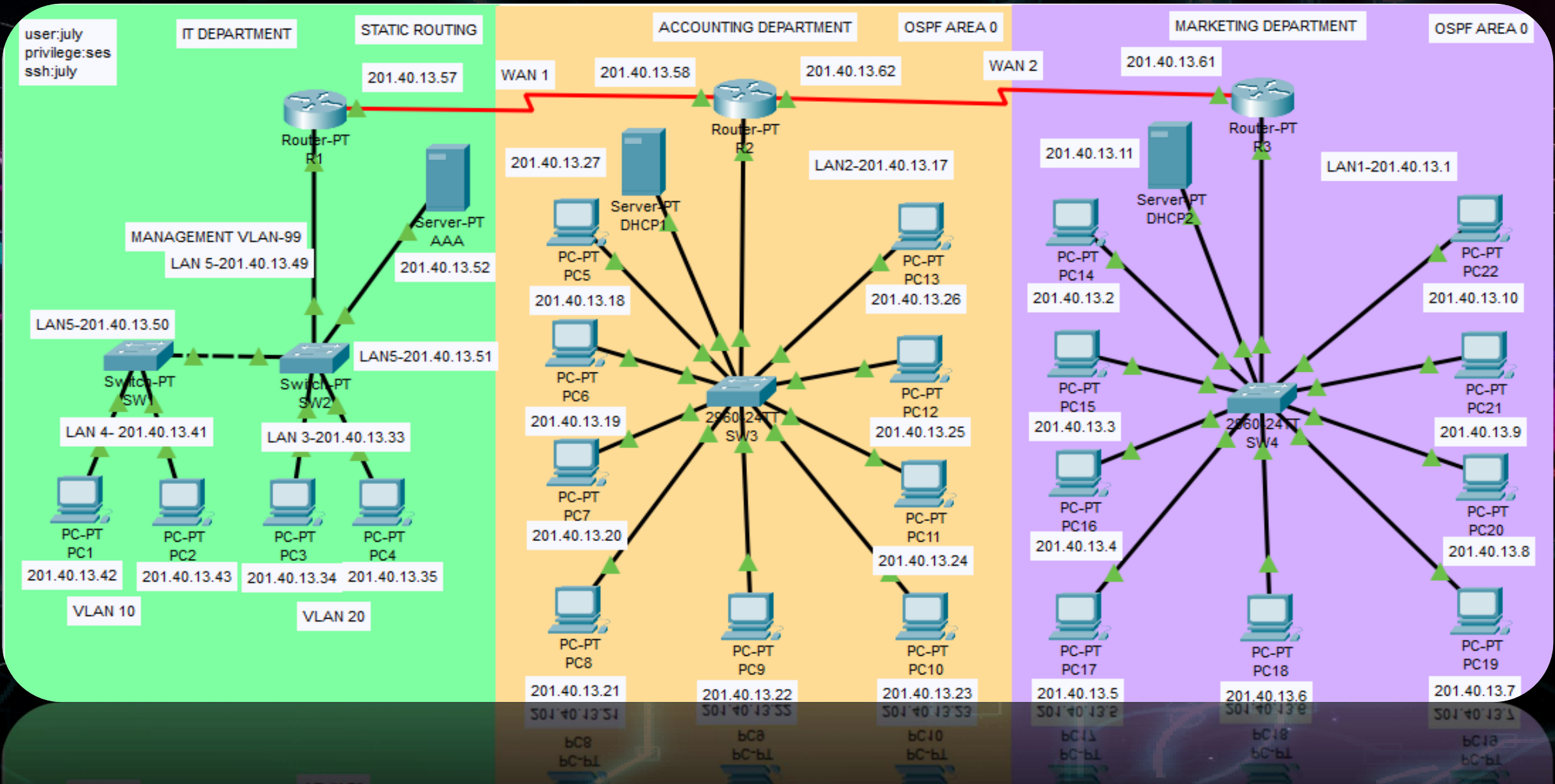
The background is a dark, deep blue or black. It is filled with a complex network of thin, glowing lines in shades of cyan, blue, and purple, resembling a fiber-optic or data network. These lines are interconnected with small square and circular nodes. A prominent, thick, multi-colored wave or ribbon sweeps across the lower half of the image, transitioning from a bright cyan on the left to a vibrant magenta and purple on the right. The overall effect is high-tech and futuristic.

# ***Implementation of Secure Hybrid Routing, VLSM, and AAA Enterprise Architecture***

# ***TOPOLOGY***

The background is a dark, textured surface with a complex pattern of glowing, wavy lines in shades of purple, blue, and green. These lines flow across the frame, creating a sense of movement and depth. Overlaid on this are faint, light-blue geometric shapes, including squares, circles, and lines, which form a network-like structure. The overall aesthetic is futuristic and mathematical, fitting the theme of topology.

# TOPOLOGY





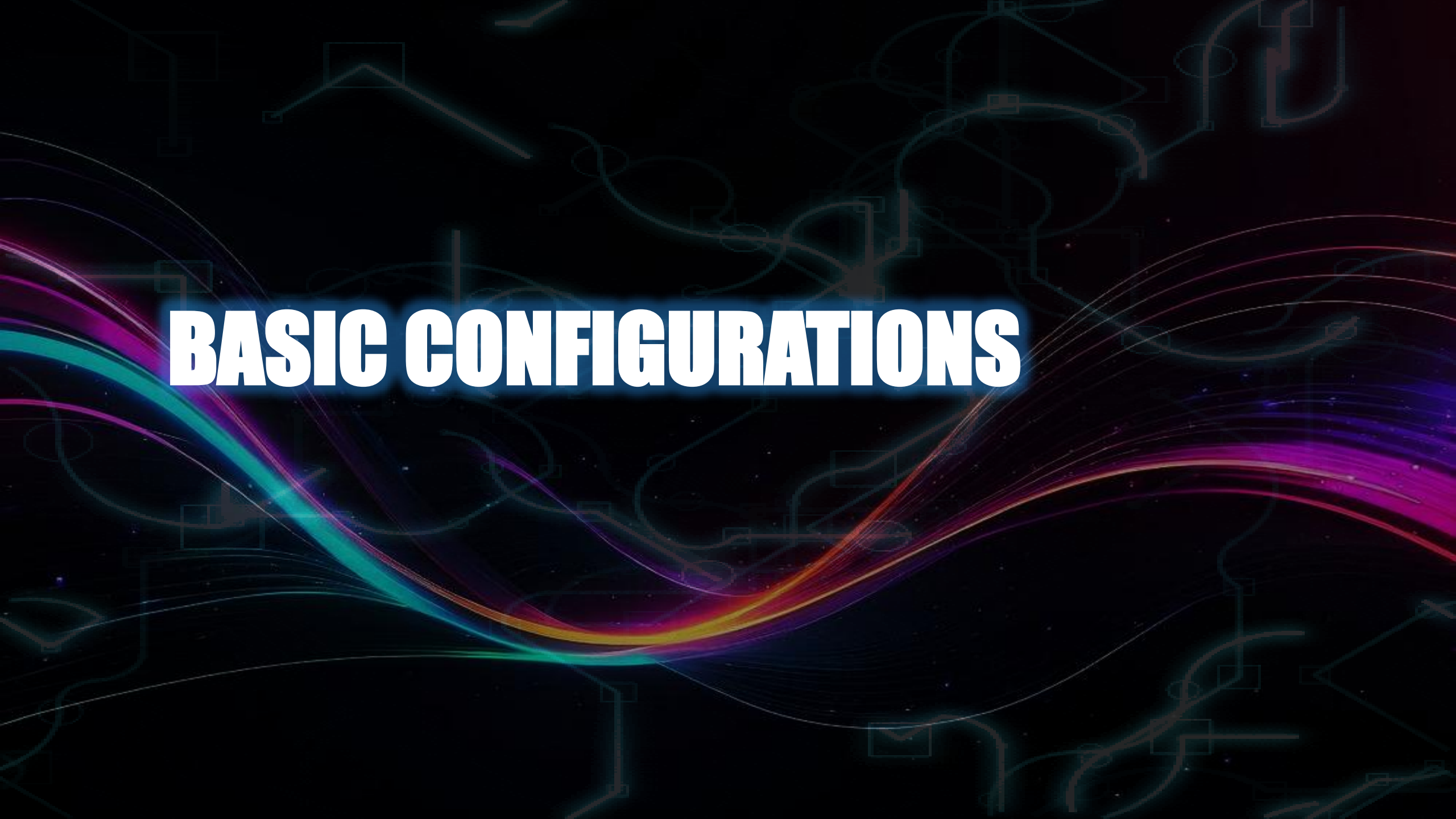
The background is a dark, textured space filled with glowing, ethereal lines and faint geometric shapes. A prominent, multi-colored wave of light, transitioning from teal to purple to yellow, curves across the lower half of the image. In the upper half, there are faint, glowing lines and shapes that resemble a complex network or circuitry. The overall effect is one of dynamic energy and futuristic technology.

***SCENARIO***

## **Interdepartmental Enterprise Network with AAA(TACACS+) and Hybrid Routing Implementation**

There are 3 Departments in the Main Office Building of Horizon Group of Companies . These are the IT Department, Accounting Department and Marketing Department. In the IT Department there are 5 hosts and 3 VLANs which are the VLAN 99 for Management , VLAN 10 for IT Security Department and VLAN 20 for Network Administration . The IT Department is configured for the following: Static Routing , ROAS or Router on A Switch and Authentication , Authorization and Accounting or known as AAA( TACACS+) for security. The Accounting Department and Marketing Department, there are a total of 11 hosts and OSPF or Open Shortest Path First and DHCP or Dynamic Host Control Protocol is implemented for the network.

# BASIC CONFIGURATIONS

The background is a dark, textured surface with a complex pattern of glowing, thin lines in shades of blue, green, and yellow. These lines form a network of interconnected nodes and paths, resembling a circuit board or a data network. The overall effect is a sense of dynamic energy and technological sophistication.



## BASIC CONFIGURATIONS FOR:

### -SWITCHES

### -ROUTERS

```
Router(config)#hostname LAN1_ROUTER
LAN1_ROUTER(config)#enable secret ses
LAN1_ROUTER(config)#line console 0
LAN1_ROUTER(config-line)#password july
LAN1_ROUTER(config-line)#login
LAN1_ROUTER(config-line)#exit
LAN1_ROUTER(config)#line vty 0 4
LAN1_ROUTER(config-line)#password july
LAN1_ROUTER(config-line)#login local
LAN1_ROUTER(config-line)#transport input ssh
LAN1_ROUTER(config-line)#exit
LAN1_ROUTER(config)#banner motd #
```

```
+-----+
| UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE AND |
| ATTACHED NETWORKS IS STRICTLY PROHIBITED.      |
| You must have explicit permission to access or |
| configure this device. All activities performed on |
| this device may be logged or monitored without further |
| notice, and the resulting logs may be used as evidence |
| in court.                                           |
| Any unauthorized use of the system is unlawful, and |
| may be subject to civil and/or criminal penalties. |
+-----+ #
```

```
LAN1_ROUTER(config)#service password-encryption
LAN1_ROUTER(config)#end
LAN1_ROUTER#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

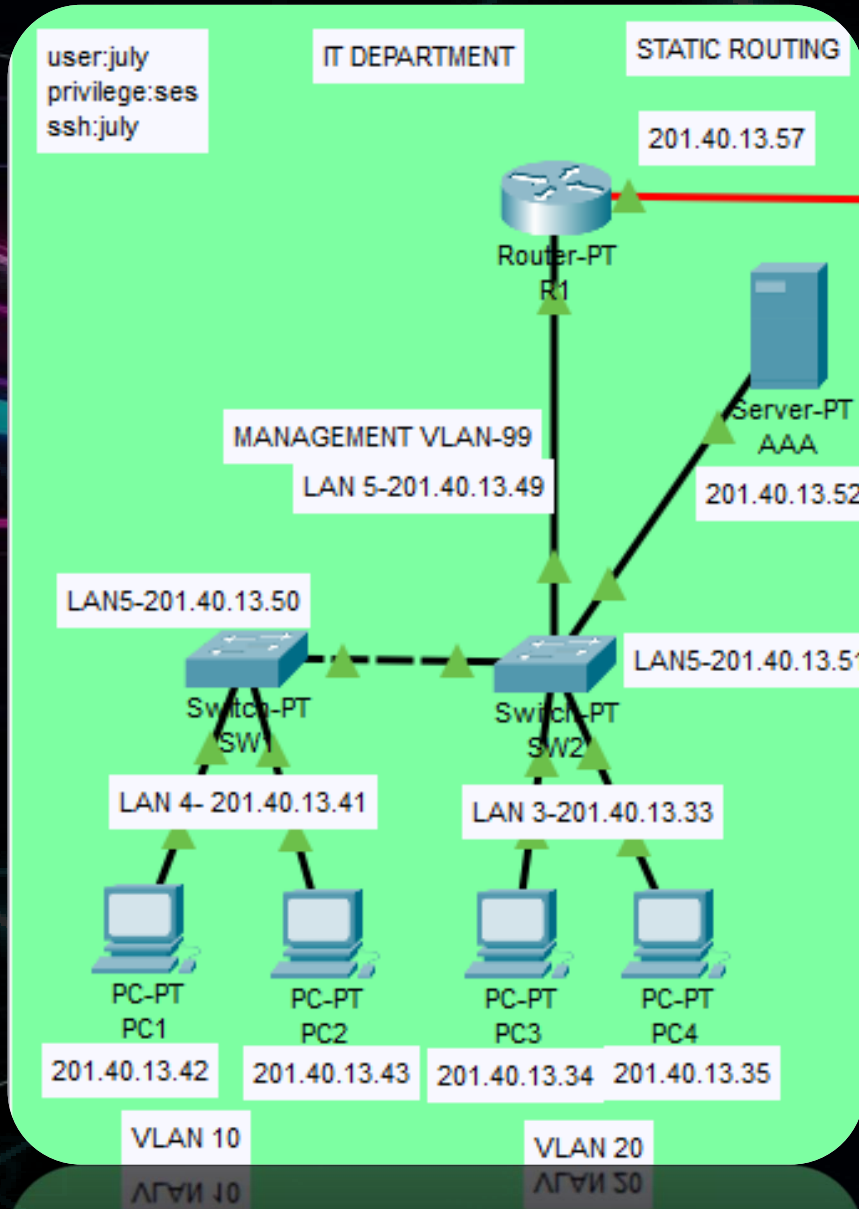
The background is a dark blue field with a complex network of glowing green lines and nodes, resembling a circuit board or data network. Overlaid on this are several vibrant, flowing streaks of light in shades of purple, magenta, and cyan. The text 'IT DEPARTMENT' is centered in a bold, white, sans-serif font with a blue glow effect.

**IT DEPARTMENT**



# TOPOLOGY

# IP ADDRESS TABLE/VLAN TABLE



| AREA  | DEVICE | PORT | IP ADD       | SUBNET MASK         | GATEWAY ADDRESS | WILDCARD MASK |
|-------|--------|------|--------------|---------------------|-----------------|---------------|
| LAN 4 | PC1    | NIC  | 201.40.13.42 | /29-255.255.255.248 | 201.40.13.41    | 0.0.0.7       |
| LAN 4 | PC2    | NIC  | 201.40.13.43 | /29-255.255.255.248 | 201.40.13.41    | 0.0.0.7       |
| LAN 3 | PC3    | NIC  | 201.40.13.34 | /29-255.255.255.248 | 201.40.13.33    | 0.0.0.7       |
| LAN 3 | PC4    | NIC  | 201.40.13.35 | /29-255.255.255.248 | 201.40.13.33    | 0.0.0.7       |
| LAN 5 | SW1    | F0/1 | 201.40.13.50 | /29-255.255.255.248 | 201.40.13.49    | 0.0.0.7       |
| LAN 5 | SW2    | F0/1 | 201.40.13.51 | /29-255.255.255.248 | 201.40.13.49    | 0.0.0.7       |
|       |        | F1/1 | 201.40.13.49 | /29-255.255.255.248 | N/A             | 0.0.0.7       |
| LAN 5 | AAA    | NIC  | 201.40.13.52 | /29-255.255.255.248 | 201.40.13.49    | 0.0.0.7       |
| WAN 1 | R1     | S2/0 | 201.40.13.57 | /30-255.255.255.252 | N/A             | 0.0.0.3       |
|       |        | F1/0 | 201.40.13.49 | /29-255.255.255.248 | N/A             | 0.0.0.7       |

| VLAN TABLE |         |           |              |                     |                 |
|------------|---------|-----------|--------------|---------------------|-----------------|
| DEVICE     | VLAN    | INTERFACE | IP ADDRESS   | SUBNET MASK         | DEFAULT GATEWAY |
| R3         | VLAN 10 | F1/0.10   | 201.40.13.41 | /29-255.255.255.248 | N/A             |
|            | VLAN 20 | F1/0.20   | 201.40.13.33 |                     | N/A             |
|            | VLAN 99 | F1/0.99   | 201.40.13.49 |                     | N/A             |
| SW1        | VLAN 10 | F2/1      | 201.40.13.42 |                     | 201.40.13.41    |
|            |         | F1/1      | 201.40.13.43 |                     | 201.40.13.41    |
| SW2        | VLAN 99 | VLAN 99   | 201.40.13.50 |                     | 201.40.13.49    |
|            | VLAN 20 | F3/1      | 201.40.13.35 |                     | 201.40.13.33    |
| SW2        | VLAN 20 | F2/1      | 201.40.13.34 |                     | 201.40.13.33    |
|            |         | VLAN 99   | 201.40.13.51 |                     | 201.40.13.49    |
| PC1        | VLAN 10 | NIC       | 201.40.13.42 |                     | 201.40.13.41    |
| PC2        |         |           | 201.40.13.43 |                     | 201.40.13.41    |
| PC3        |         |           | 201.40.13.34 |                     | 201.40.13.33    |
| PC4        |         |           | 201.40.13.35 |                     | 201.40.13.33    |
| AAA SERVER | VLAN 99 |           | 201.40.13.52 |                     | 201.40.13.49    |

# CONFIGURATIONS

The background is a dark, deep blue or black field. It is populated with a complex network of glowing, thin lines in shades of cyan, magenta, and purple. These lines flow and curve across the frame, creating a sense of dynamic movement. Interspersed among these lines are numerous small, faint geometric shapes, including squares, circles, and triangles, some of which appear to be part of a larger, more intricate pattern or circuitry. The overall aesthetic is futuristic and technological.

# R1 STATIC CONFIGURATION

**R1# conf t**

**R1(config)# ip route 201.40.13.16 255.255.255.240 s2/0**

**R1(config)# ip route 201.40.13.0 255.255.255.240 s2/0**

**R1(config)# ip route 201.40.13.60 255.255.255.252 s2/0**



# SW1 INTER-VLAN CONFIGURATION

**SW1(config)#vlan 10**

**SW1(config-vlan)#vlan 20**

**SW1(config-vlan)#vlan 99**

**SW1(config-vlan)#exit**

**SW1(config)#interface vlan 99**

**SW1(config-if)#ip address 201.40.13.50 255.255.255.248**

**SW1(config-if)#exit**

**SW1(config)#ip default-gateway 201.40.13.49**

**SW1(config)#interface range FastEthernet1/1, FastEthernet2/1**

**SW1(config-if-range)#switchport mode access**

**SW1(config-if-range)#switchport access vlan 10**

**SW1(config-if-range)#no shutdown**

**SW1(config-if-range)#exit**

**SW1(config)#interface FastEthernet0/1**

**SW1(config-if)#switchport mode trunk**

# SW2 INTER-VLAN CONFIGURATION

```
SW2(config)# vlan 10
```

```
SW2(config-vlan)# vlan 20
```

```
SW2 (config-vlan)# vlan 99
```

```
SW2(config-vlan)# exit
```

```
SW2(config)# interface vlan 99
```

```
SW2(config-if)# ip address 201.40.13.51 255.255.255.248
```

```
SW2(config-if)# no shutdown
```

```
SW2(config-if)# exit
```

```
SW2(config)# ip default-gateway 201.40.13.49
```

```
SW2(config)# interface range fastethernet3/1, fastethernet2/1,
```

```
SW2(config-if-range)# switchport mode access
```

```
SW2(config-if-range)# switchport access vlan 20
```

```
SW2(config-if-range)# exit
```

```
SW2(config)# interface range fastethernet1/1, fastethernet4/1
```

```
SW2(config-if-range)# switchport mode trunk
```

```
SW2(config)#int f4/1
```

```
SW2(config-if)#switchport mode access
```

```
SW2(config-if)#switchport access vlan 99
```

```
SW2(config-if)#
```

# AAA SERVER CONFIGURATION

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

AAA

Service

☒ On
 ☐ Off

Radius Port

1645

Network Configuration

Client Name

Client IP

Secret

ServerType

Radius

|   | Client Name | Client IP    | Server Type | Key   |                   |
|---|-------------|--------------|-------------|-------|-------------------|
| 1 | R1          | 201.40.13.49 | Tacacs      | mykey | <div>Add</div>    |
|   |             |              |             |       | <div>Save</div>   |
|   |             |              |             |       | <div>Remove</div> |

User Setup

Username

Password

|   | Username | Password |                |
|---|----------|----------|----------------|
| 1 | ses      | ses      | <div>Add</div> |
|   |          |          | <div>Add</div> |



# R1 AAA CONFIGURATION

**R1# configure terminal**

**R1(config)# aaa new-model**

**R1(config)# aaa authentication login myauth group tacacs+ local**

**R1(config)# tacacs-server host 201.40.13.52 key mykey**

**R1(config)#access-list 101 deny tcp 201.40.13.16 0.0.0.15 any eq telnet**

**R1(config)#access-list 101 deny tcp 201.40.13.0 0.0.0.15 any eq telnet**

**R1(config)#access-list 101 permit tcp any any eq telnet**

# **R1 AAA CONFIGURATION**

**R1(config)# line vty 0 4**

**R1(config-line)# login authentication myauth**

**R1(config-line)#access-class 101 in**

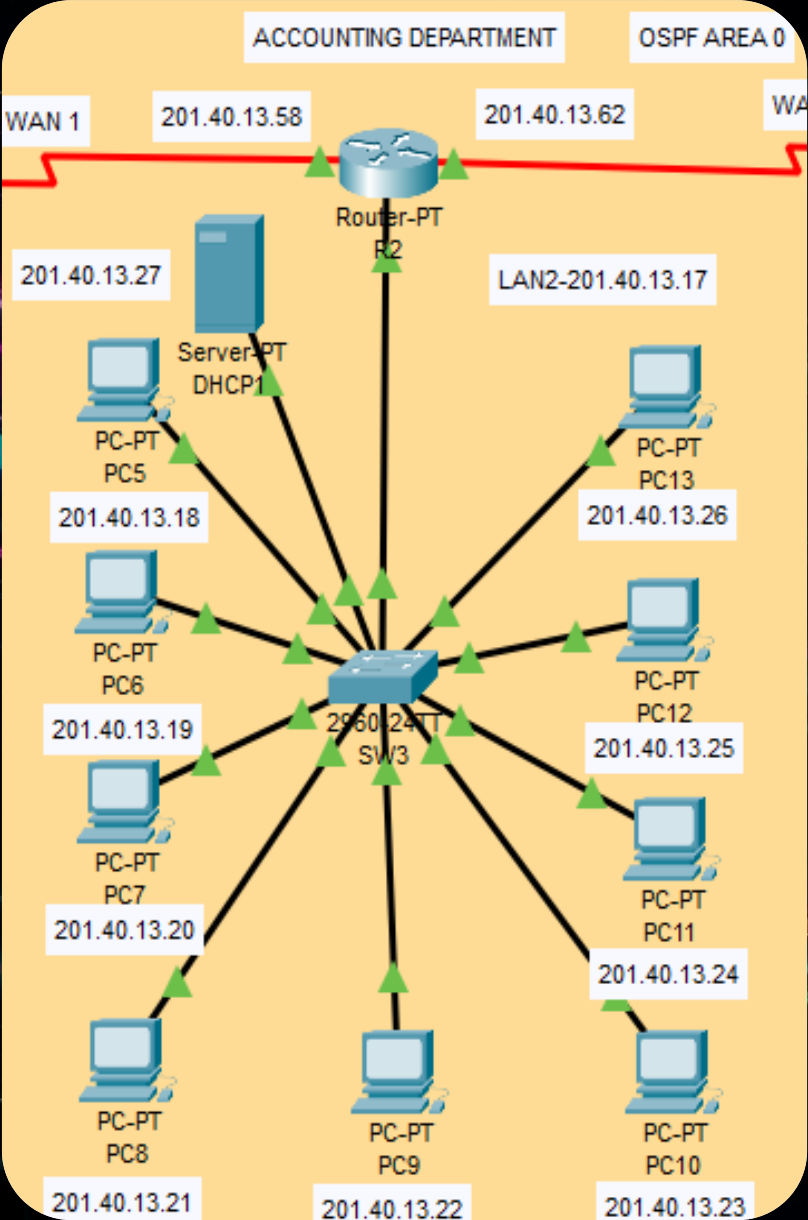
**R1(config-line)#exit**

# ACCOUNTING DEPARTMENT

The background is a dark, textured surface with a complex network of glowing, thin lines in shades of blue, green, and yellow. These lines form various geometric shapes, including squares, circles, and triangles, some of which are interconnected. A prominent, thick, wavy band of vibrant colors (red, orange, yellow, green, and blue) curves across the lower half of the image, creating a sense of motion and energy.



# TOPOLOGY



# IP ADDRESS TABLE/VLAN TABLE

| AREA  | DEVICE | PORT | IP ADD       | SUBNET MASK         | GATEWAY ADDRESS | WILDCARD MASK |
|-------|--------|------|--------------|---------------------|-----------------|---------------|
| WAN 2 | R3     | F0/0 | 201.40.13.1  | /28-255.255.255.240 | N/A             | 0.0.0.15      |
|       |        | S2/0 | 201.40.13.61 | /30-255.255.255.252 | N/A             | 0.0.0.3       |
| LAN 2 | PC5    | NIC  | 201.40.13.18 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |
| LAN 2 | PC6    | NIC  | 201.40.13.19 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |
| LAN 2 | PC7    | NIC  | 201.40.13.20 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |
| LAN 2 | PC8    | NIC  | 201.40.13.21 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |
| LAN 2 | PC9    | NIC  | 201.40.13.22 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |
| LAN 2 | PC10   | NIC  | 201.40.13.23 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |
| LAN 2 | PC11   | NIC  | 201.40.13.24 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |
| LAN 2 | PC12   | NIC  | 201.40.13.25 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |
| LAN 2 | PC13   | NIC  | 201.40.13.26 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |
| LAN 2 | DHCP1  | NIC  | 201.40.13.27 | /28-255.255.255.240 | 201.40.13.17    | 0.0.0.15      |

**DHCP POOL: 201.40.13.18 - 201.40.13.30**

# CONFIGURATIONS

The background is a dark, deep blue or black space. It is filled with a complex network of glowing, thin lines in shades of cyan, magenta, and purple. These lines flow and curve across the frame, creating a sense of dynamic movement. Interspersed among these lines are various faint, light-colored geometric shapes, including squares, circles, and triangles, some of which appear to be part of a larger, interconnected grid or circuit-like structure. The overall effect is one of high-tech, digital, or futuristic aesthetics.

# R2 OSPF CONFIGURATION

**R2(config)# router ospf 10**

**R2(config-router)# network 201.40.13.16 0.0.0.15 area 0**

**R2(config-router)# network 201.40.13.60 0.0.0.3 area 0**

**R2(config-router)# passive-interface FastEthernet0/0**



# R2 STATIC CONFIGURATION

```
R2(config)# ip route 201.40.13.48 255.255.255.248 s2/0
```

```
R2(config)# ip route 201.40.13.32 255.255.255.248 s2/0
```

```
R2(config)# ip route 201.40.13.40 255.255.255.248 s2/0
```

# DHCP1 SERVER CONFIGURATION

DHCP1

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface

FastEthernet0

Service

☒ On

☐ Off

Pool Name

serverPool

Default Gateway

201.40.13.17

DNS Server

0.0.0.0

Start IP Address :

201

40

13

18

Subnet Mask:

255

255

255

240

Maximum Number of Users :

14

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

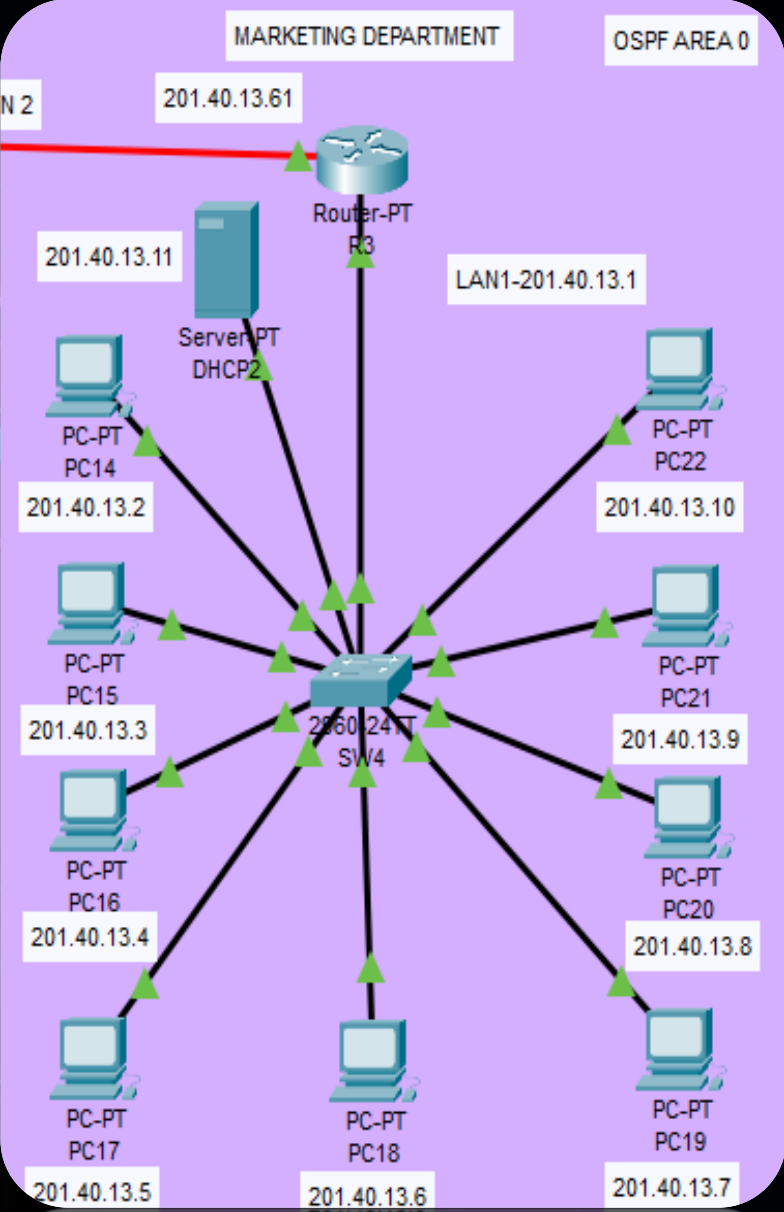
| Pool Name  | Default Gateway | DNS Server | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|------------|-----------------|------------|------------------|-------------|----------|-------------|-------------|
| serverPool | 201.40.1...     | 0.0.0.0    | 201.40.1...      | 255.255.... | 14       | 0.0.0.0     | 0.0.0.0     |

# MARKETING DEPARTMENT

The background is a dark, textured surface with a complex network of glowing, thin lines in shades of blue, green, and yellow. These lines form various geometric shapes, including squares, circles, and triangles, some of which are interconnected. A prominent, thick, wavy line in vibrant purple and magenta sweeps across the lower half of the image, adding a sense of motion and energy.



# TOPOLOGY



# IP ADDRESS TABLE/VLAN TABLE

| AREA  | DEVICE | PORT | IP ADD       | SUBNET MASK         | GATEWAY ADDRESS | WILDCARD MASK |
|-------|--------|------|--------------|---------------------|-----------------|---------------|
| LAN 1 | PC14   | NIC  | 201.40.13.2  | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | PC15   | NIC  | 201.40.13.3  | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | PC16   | NIC  | 201.40.13.4  | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | PC17   | NIC  | 201.40.13.5  | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | PC18   | NIC  | 201.40.13.6  | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | PC19   | NIC  | 201.40.13.7  | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | PC20   | NIC  | 201.40.13.8  | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | PC21   | NIC  | 201.40.13.9  | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | PC22   | NIC  | 201.40.13.10 | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | DHCP2  | NIC  | 201.40.13.11 | /28-255.255.255.240 | 201.40.13.1     | 0.0.0.15      |
| LAN 1 | SW4    | F0/1 | 201.40.13.1  | /28-255.255.255.240 | N/A             | N/A           |
| WAN 2 | R3     | F0/0 | 201.40.13.1  | /28-255.255.255.240 | N/A             | 0.0.0.15      |
|       |        | S2/0 | 201.40.13.61 | /30-255.255.255.252 | N/A             | 0.0.0.3       |

**DHCP POOL: 201.40.13.2 - 201.40.13.14**

# CONFIGURATIONS

The background is a dark, textured space filled with glowing, ethereal lines and shapes. A prominent, multi-colored wave of light, transitioning from teal to purple to yellow, curves across the lower half of the image. In the upper half, there are faint, glowing geometric patterns, including squares and circles, some of which are interconnected by thin lines, suggesting a network or a complex system. The overall aesthetic is futuristic and technological.

# **R3 OSPF CONFIGURATION**

**R3(config)# router ospf 10**

**R3(config-router)# network 201.40.13.0 0.0.0.15 area 0**

**R3(config-router)# network 201.40.13.60 0.0.0.3 area 0**

**R3(config-router)# passive-interface FastEthernet0/0**



# R3 STATIC CONFIGURATION

**R3(config)# ip route 201.40.13.48 255.255.255.248 s2/0**

**R3(config)# ip route 201.40.13.32 255.255.255.248 s2/0**

**R3(config)# ip route 201.40.13.40 255.255.255.248 s2/0**

**R3(config)# ip route 201.40.13.56 255.255.255.252 s2/0**